

Willow Hughes

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About Me

Machine learning researcher and developer with a passion for applying AI to complex, real-world problems.

Education

MS	Western Washington University Computer Science 4.00 GPA	Bellingham, WA Jan 2026 – Mar 2027
BS	Western Washington University Computer Science, Pre-Masters and Honors Curriculum 3.84 GPA	Bellingham, WA Jan 2023 – Dec 2025

Experience

AI & Data Analytics Intern Pacific Northwest National Laboratory National Security Directorate	Seattle, WA Summer 2026
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- Incoming intern working on AI Assurance

Deep Learning Research Assistant Hutchinson Machine Learning Research Group	Bellingham, WA Dec 2025 – Present
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- Developing deep learning segmentation models in PyTorch for LiDAR-based Maya site detection, reaching 0.74 IoU on structures, collaborating with archaeologists to address domain-specific challenges
- Re-engineered the data pipeline on a fragile shared NFS (lossless 3TB to 500GB compression, staged caching, multiworker generation), cutting data generation from 24h to 5h and epoch time from 90m to 30m
- Participating in a weekly ML reading group on foundational LLM research

Teaching Assistant, Object-Oriented Design Western Washington University	Bellingham, WA Mar 2026 – June 2026
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Undergraduate Research Assistant Western Washington University	Bellingham, WA Oct 2024 – Dec 2025
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- First author on research paper investigating Test-Driven Development in CS education; submitted to ITiCSE 2026 conference

Software Engineering Intern Premera Blue Cross	Seattle, WA Jun 2024 – Aug 2024
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- Led the cloud migration of a C#/SQL web application to Azure, coordinating with Microsoft engineers to assist with strategy
- Built CI/CD (YAML) pipelines and IaC (Bicep) templates for core services, reducing deployment time by >40%
- Implemented authentication and authorization with Azure Entra ID

Projects

Conversational AI Spanish Tutor (Python, TypeScript, React, Flask, REST APIs)

- Building a real-time voice AI system with a Python/Flask REST API backend and React frontend, integrating STT, LLM, and TTS services
- Optimizing the end-to-end inference pipeline to achieve sub-second voice response latency

Multithreaded Image Processing Application (C, POSIX Threads)

- Developed a C program utilizing POSIX threads to implement a Laplacian edge detection algorithm on PPM images
- Engineered an efficient image filtering pipeline, achieving up to ~70% runtime reduction through optimized thread synchronization and workload distribution

Skills

Languages: Python, Java, C, C#, SQL, TypeScript, JavaScript

Tools & Frameworks: Git, Docker, Linux/Unix, Bash/Shell, CI/CD, Azure, AWS, GCP, React, REST APIs, PyTorch, NumPy

Involvement

WWU's First Hackathon (Organizing Team)	2026 – Present
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- Consulting with local nonprofits to source real-world projects for student teams to build during the event

Orchestra & Band Involvement	2013 – Present
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